

Racconbot's functional enhancement

- Enhancing Raccoonbot's ability

1. Project Overview.

Extending RaccoonBot pipeline by adding multiple objects, tasks, and instructions while improving OpenVLA action mapping and adding YOLO for real robot

2. Extension

< New objects >

- Color : red, blue, green, yellow
- Shape: cylinder, cube, sphere, box

< New instructions >

- grasp : "pick up the {color} cylinder"
- lift : "raise up the {color} cylinder"
- push : "move the {color} cylinder forward"
- pick-and-place : "pick up the {color} cylinder and place it"
- stack : "place the {color} cylinder on top of another"
- swap : "switch the {color} cylinder position"

< Dataset >

- 400 MuJoCo demonstration generated
- RLDS/TFDS rebuild
- LoRa fine-tuned

3. Improvement

< 7D-to-4DOF Action Mapping >

- max_delta_xyz :: 0.01 -> 0.015
- delta_scale : 1.0 -> 1.2
- max_retries : 3 -> 5
- Effect : make it faster and more reliable robot motion

< Timing Log >

- Added [TIMING] inference_time log to openvla_server.py
- Inference time : 0.21~0.27s per step

4. Additional

< YOLO Webcam Color Detection >

- Detect target color from webcam using color range
- Automatically chooses target color based on detected color

< Limitation >

- Sim to Real gap : Real robot moves based on OpenVla, not using camera
- Future work : set real environment based on the camera

5. Discussion

What makes the system faster, clearer and more reliable?

- faster : Increasing delta_scale and max_delta_xyz allow the robot to reach target much faster
- Clearer : Timing logs provide clear visibility into inference bottlenecks
- More reliable : Increasing max_retries reduces IK failure rate
- Future work: Keep working on the limitation about Sim to Real gap in real robot control